

$$V_R + V_L(t) = 0$$

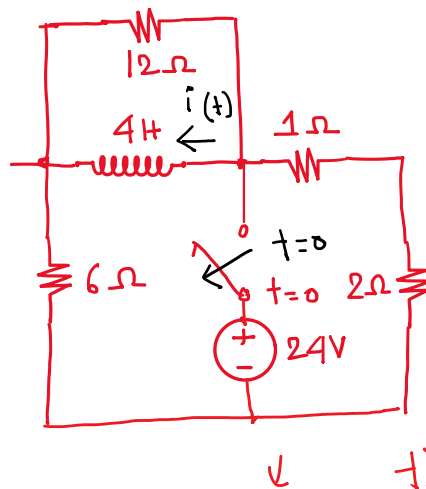
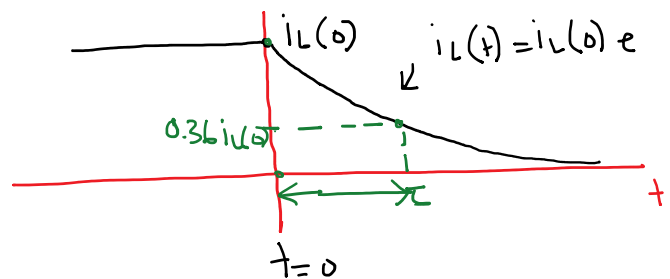
$$\Rightarrow i_L(t)R + L \frac{di_L(t)}{dt} = 0$$

$$\frac{di_L(t)}{dt} + \frac{R}{L} i_L(t) = 0 \rightarrow RL$$

$$\frac{dV_L(t)}{dt} + \frac{V_L(t)}{R/L} = 0 \rightarrow RC$$

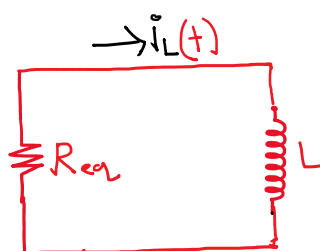
$$-\frac{1}{\tau}$$

$$i_L(t) = i_L(0) e^{-\frac{t}{\tau}} \quad \tau = \frac{L}{R}$$



Switch was closed for long...
at $t=0$, it's open.

Find $i(t)$ for all values of t : $-\infty < t < \infty$



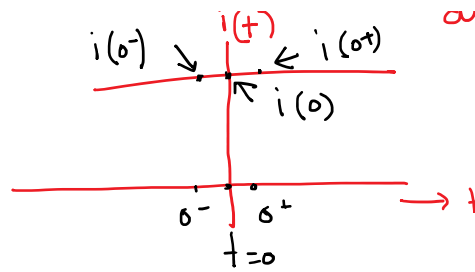
$$i_L(t) = i_L(0) e^{-\frac{t}{\tau}}$$

τ	For $t \geq 0$
$i_L(0)$	$t < 0$
	4A

1. Current through inductor doesn't change abruptly: $V_L(t) = L \frac{di_L(t)}{dt}$



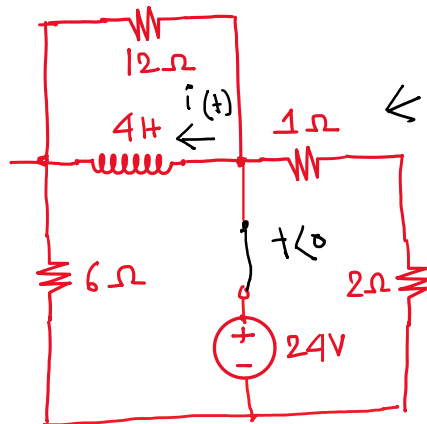
$$\frac{5-0}{\frac{1}{2} - 1} \rightarrow \infty \uparrow$$



$$i(0^-) = i(0) = i(0^+)$$

$i(0) = i(0^-)$, How to find $i(0^-)$??

determine $i_L(0)$ for $t < 0$.

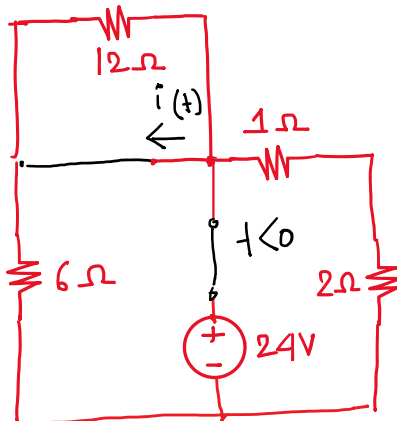


CKT for $t < 0$:

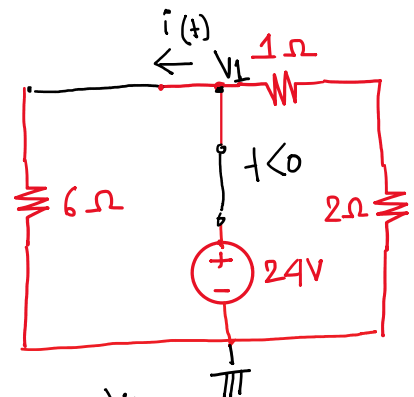
□ Inductor behaves like a short ckt to DC signal

$$V_L(t) = L \frac{di(t)}{dt} = 0$$

↓



12 ohm Resistor shorted out



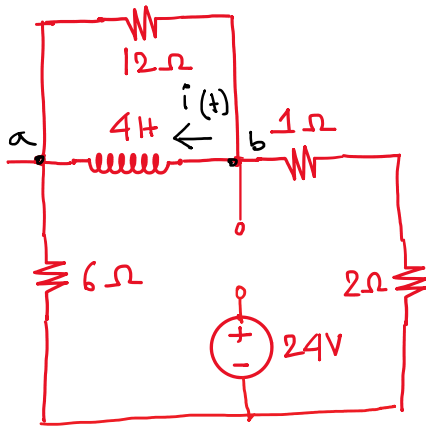
$$V_1 = 24V, i(t) = \frac{V_1}{6} = \frac{24}{6} = 4A$$

$i(t) = 4A$ [current through the inductor for $t < 0$]

$$i(0) = i(0^-) = 4A$$

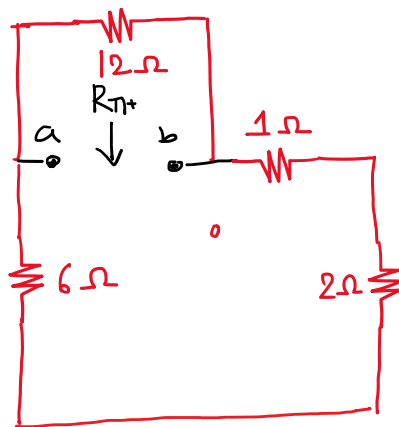


for $\tau = \frac{L}{R_{eq}}$ consider the CKT for

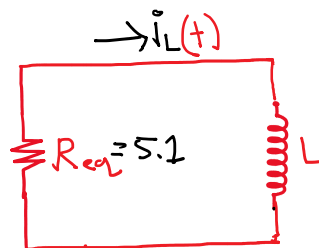


for $\tau = \frac{L}{R_{eq}}$ Consider the ckt for $t > 0$

- Steps:
1. Remove the inductor.
 2. Look through both the two nodes of the removed inductor site to obtain R_{TH}



$$\begin{aligned}
 R_{eq} = R_{TH} &= 12 \parallel (1 + 2 + 6) \\
 &= 12 \parallel 9 \\
 &= \frac{12 \times 9}{12 + 9} = \frac{36}{7} \\
 &= 5.1 \Omega
 \end{aligned}$$



$$\begin{aligned}
 \tau &= \frac{L}{5.1} = \frac{4}{5.1} \text{ sec} \\
 &= 0.7 \text{ sec} \\
 i_L(t) &= 4 e^{-\frac{t}{0.7}}
 \end{aligned}$$